CSM
Fall 2026

EE 64
Worksheet 7

1. Phasor and Impedance Basics

(a) What is Euler's identity?

Solution: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$

(b) Check that $\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}$ holds using Euler's identity.

Solution: $e^{-j\theta} = \cos(\theta) - j\sin(\theta)$ because cosine is an even function while sine is odd. Thus, $e^{-j\theta} + e^{j\theta} = 2\cos(\theta) \implies \cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}$

(c) What is the phasor form of the signal?

$$v(t) = V_0\cos(\omega t + \phi_v) \tag{1}$$

Solution: The coefficient of the $e^{j\omega t}$ term in equation (1) is called the phasor form of this signal:

$$\widetilde{V} = V_0 e^{j\phi_v}$$

(d) What is the magnitude and phase of the expression $8e^{j\frac{\pi}{3}}$?

Solution: Magnitude is 8. Phase is $\frac{\pi}{3}$.

(e) Transform the signal $12\cos\left(\omega t - \frac{3\pi}{4}\right)$ to phasor domain, in the form $Ae^{jB\pi}$.

Solution:

The phasor form of the signal is $12e^{j-\frac{3}{4}\pi}$.

(f) What is the impedance of a capacitor C?

Solution: $\frac{1}{j\omega C}$

(g) What is the impedance of an inductor L?

Solution: $j\omega L$

(h) What is the impedance of a resistor R?

Solution: R

(i) What is the impedance of an resistor R and a capacitor C in parallel?

Solution: $\frac{R}{1+j\omega RC}$

(j) What is the impedance of an inductor L and a capacitor C in parallel?

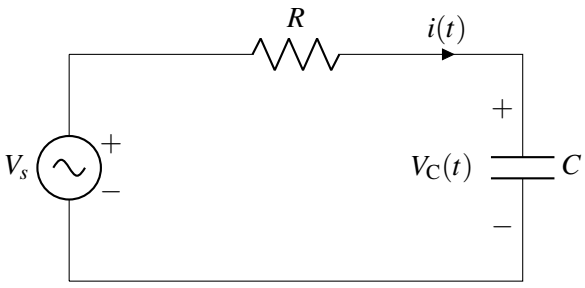
Solution: $\frac{j\omega L}{1-\omega^2 LC}$

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2. Phasor Analysis

Equipped with the phasor representation, let's see how using it can greatly simplify alternating current (AC) circuit analysis, assuming that the voltages and currents vary with time as sinusoids.

The phasor analysis method consists of five steps. We'll walk through it in the RC circuit below:



The voltage source is given by

$$V_s(t) = 8\cos\left(\omega t - \frac{\pi}{4}\right), \tag{2}$$

with $\omega = 10^3$ rad/s, $R = 1$ kΩ, and $C = 1$ μF.

Our goal is to obtain a solution for $V_C(t)$ and $i(t)$ with the sinusoidal voltage source $V_s(t)$.

(a) **Step 1: Convert $v(t)$ as a sum of complex exponentials.**

All voltages and currents with known sinusoidal functions should be expressed in the standard exponential format. **Convert $V_s(t)$ into an exponential and write down its phasor representation $\widetilde{V}_s$.**

Solution: We can extract the phasor representation of $V_s(t)$ by writing it in the form $V_s(t) = \mathrm{Re}(\widetilde{V}_s e^{j\omega t})$:

$$\begin{aligned} V_s(t) &= 8\cos\left(\omega t - \frac{\pi}{4}\right) \\ &= \mathrm{Re}(8e^{j(-\frac{\pi}{4})}e^{j\omega t}) \\ \Longrightarrow \widetilde{V}_s &= 8e^{j(-\frac{\pi}{4})} \end{aligned}$$

Alternatively, we can expand $V_s(t)$ into the sum of complex exponentials and pattern matching to find the phasor.

The two formulas below will help us convert a sinusoid into a sum of exponentials:

$$\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta} \quad \text{and} \quad \sin(\theta) = \frac{1}{2j}e^{j\theta} - \frac{1}{2j}e^{-j\theta}$$

Using the fact that $\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}$,

$$\begin{aligned} V_s(t) &= 8\left(\frac{1}{2}e^{j(\omega t - \frac{\pi}{4})} + \frac{1}{2}e^{-j(\omega t - \frac{\pi}{4})}\right) = \frac{1}{2}(e^{j(\omega t - \frac{\pi}{4})} + 8e^{-j(\omega t - \frac{\pi}{4})}) \\ &= \frac{1}{2}(8e^{-j\frac{\pi}{4}}e^{j\omega t} + 8e^{j\frac{\pi}{4}}e^{-j\omega t}) \end{aligned}$$

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Thus, the phasor is given by

$$\widetilde{V}_s = 8e^{-j\frac{\pi}{4}}$$

(b) **Step 2: Transform circuits elements to phasor domain impedances.**
The voltage source is represented by its phasor $\widetilde{V}_s$. The current $i(t)$ is related to its phasor counterpart $\widetilde{I}$. **What are the phasor representations of R and C?**
Solution:

$$Z_R = R \tag{3}$$

$$Z_C = \frac{1}{j\omega C} \tag{4}$$

(c) **Step 3: Cast KCL and/or KVL equations in phasor domain.**
Use Kirchhoff's laws to write down a loop equation that relates all phasors in Step 2.
Solution: By KCL, the current through the resistor is equal to the current through the capacitor.

$$i(t) = i_R(t) = i_C(t) \rightarrow \widetilde{I} = \widetilde{I}_R = \widetilde{I}_C$$

By KVL, the sum of the voltages across the resistor and capacitor is equal to that of the source.

$$V_s(t) = V_C(t) + V_R(t) \rightarrow \widetilde{V}_s = \widetilde{V}_c + \widetilde{V}_R$$

Substituting the impedances in, we get:

$$\widetilde{V}_s = \widetilde{I}\frac{1}{j\omega C} + \widetilde{I}R = \widetilde{I}(R + \frac{1}{j\omega C})$$

(d) **Step 4: Solve for unknown variables**
Solve the equation you derived in Step 3 for $\widetilde{I}$ and $\widetilde{V_C}$. What is the polar form of $\widetilde{I}$ ($Ae^{i\theta}$, where A is a positive real number)?
Solution: We first solve for the current $\widetilde{I}$:

$$\widetilde{I} = \frac{\widetilde{V}_s}{R + \frac{1}{j\omega C}}$$

We plug in for $\omega = 10^3, R = 10^3\Omega, C = 10^{-6}\mu$F, $\widetilde{V}_s = 8e^{-j\frac{\pi}{4}}$:

$$\widetilde{I} = \frac{8e^{-j\frac{\pi}{4}}}{10^3 + \frac{1}{j10^3 10^{-6}}} = \frac{8e^{-j\frac{\pi}{4}}}{10^3 - 10^3 j} = \frac{8e^{-j\frac{\pi}{4}}}{10^3\sqrt{2}e^{-j\frac{\pi}{4}.}} = \frac{4\sqrt{2}}{10^3}\text{ A} = 4\sqrt{2}\text{ mA}$$

In the previous line, the third to the fourth equality comes from the plotting the denominator on the complex plane and converting to polar coordinates by taking the magnitude and the phase with respect to the positive real axis.
To solve for $\widetilde{V}_C$ we use the voltage-current relationship of a capacitor in phasor domain (as we derived in a previous problem): $\widetilde{V} = \widetilde{I}\frac{1}{j\omega C}$.

$$\widetilde{V}_C = \frac{1}{10^{-3}j}\frac{4\sqrt{2}}{10^3} = 4\sqrt{2}e^{-j(\frac{\pi}{2})}\text{ V}$$

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(e) **Step 5: Transform solutions back to time domain**

To return to time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart. What is $i(t)$ and $V_C(t)$? What is the phase difference between $i(t)$ and $V_C(t)$?

Solution: From the previous part, we now have the phasor representations of current and voltage of the capacitor:

$$\widetilde{I} = 4\sqrt{2}$$
$$\widetilde{V}_C = 4\sqrt{2}e^{-j(\frac{\pi}{2})}$$

Remember that phasors encode the magnitude and phase of a sinusoid, removing the time-dependence by assuming that we are operating with a sinusoid with the given fixed frequency ω. Therefore, to convert back, we can just read off the magnitude and phase of the current and voltage phasors.

$$\widetilde{I} = \frac{4\sqrt{2}}{10^3} = \frac{4\sqrt{2}}{10^3}e^{j(0)} \implies A = \frac{4\sqrt{2}}{10^3},\ \phi = 0$$
$$\implies i(t) = A\cos(\omega t + \phi) = \frac{4\sqrt{2}}{10^3}\cos(\omega t + 0) = 4\sqrt{2}\cos(\omega t)\ \text{mA}$$

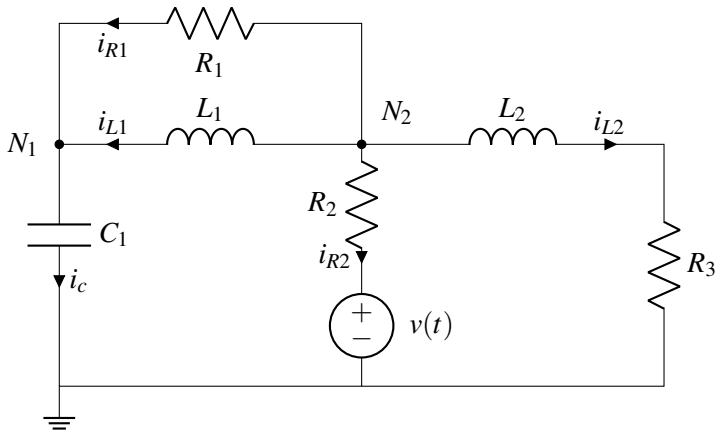
$$\widetilde{V}_C = 4\sqrt{2}e^{-j(\frac{\pi}{2})} \implies A = 4\sqrt{2},\ \phi = -\frac{\pi}{2}$$
$$\implies V_C(t) = A\cos(\omega t + \phi) = 4\sqrt{2}\cos\left(\omega t - \frac{\pi}{2}\right) = 4\sqrt{2}\cos\left(\omega t - \frac{\pi}{2}\right)\ \text{V}$$

The phase difference between the two, with respect to $i(t)$ is $-\frac{\pi}{2}$.

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3. Phasor-domain circuit analysis

The analysis techniques you learned previously for resistive circuits are equally applicable for analyzing AC circuits (circuits driven by sinusoidal inputs) in the phasor domain. In this problem, we will walk you through the steps with a concrete example. Consider the circuit below.



The components in this circuit are given by:

Voltage source:

$$v(t) = 20\cos\left(50t - \frac{\pi}{3}\right)$$

Resistors:

$$R_1 = 8\Omega, \quad R_2 = 8\Omega, \quad R_3 = 8\Omega$$

Inductors:

$$L_1 = 40\text{ mH}, \quad L_2 = 40\text{ mH}$$

Capacitor:

$$C_1 = 5\text{ mF}$$

(a) To begin with, transform the given circuit to the phasor domain.
Solution:

$$Z_L = j\omega L = j50 \times 40 \times 10^{-3} = j2\ \Omega$$
$$Z_C = \frac{1}{j\omega C} = \frac{1}{j50 \times 5 \times 10^{-3}} = -j4\ \Omega$$
$$\widetilde{v} = |v|e^{j\angle v} = 20e^{-j\frac{\pi}{3}}$$

(b) Write out KCL for node N_1 and N_2 in the phasor domain.
Solution:

At node 1

$$i_{L1} + i_{R1} = i_c$$

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At node 2

$$i_{R1} + i_{L1} + i_{R2} + i_{L2} = 0$$

(c) Use KVL to express the currents in terms of node voltages in the phasor domain. The node voltages $\widetilde{V}_1$ and $\widetilde{V}_2$ are the voltage drops from N_1 and N_2 to the ground.

Solution: We have

$$\frac{\widetilde{V}_2 - \widetilde{V}_1}{Z_L} + \frac{\widetilde{V}_2 - \widetilde{V}_1}{R} = \frac{\widetilde{V}_1}{Z_C}$$
$$\frac{\widetilde{V}_2 - \widetilde{V}_1}{R} + \frac{\widetilde{V}_2 - \widetilde{V}_1}{Z_L} + \frac{\widetilde{V}_2 - \widetilde{v}}{R} + \frac{\widetilde{V}_2}{R + Z_L} = 0$$

Plugging in values from part (a), we get

$$\frac{\widetilde{V}_2 - \widetilde{V}_1}{j2} + \frac{\widetilde{V}_2 - \widetilde{V}_1}{8} = \frac{\widetilde{V}_1}{-j4}$$
$$\frac{\widetilde{V}_2 - \widetilde{V}_1}{8} + \frac{\widetilde{V}_2 - \widetilde{V}_1}{j2} + \frac{\widetilde{V}_2 - 20e^{-j\frac{\pi}{3}}}{8} + \frac{\widetilde{V}_2}{8 + j2} = 0$$

For future parts, we want the denominators of each current to be either purely real or purely imaginary. To put i_{L2} in this form, we can manipulate the expression by multiplying the denominator by its conjugate:

$$\frac{\widetilde{V}_2}{8 + j2}\left(\frac{8 - j2}{8 - j2}\right) = \frac{\widetilde{V}_2(8 - j2)}{8^2 - (-4)} = \frac{\widetilde{V}_2(8 - j2)}{68}$$

Our final KCL equations at nodes N_1 and N_2 are

$$\frac{\widetilde{V}_2 - \widetilde{V}_1}{j2} + \frac{\widetilde{V}_2 - \widetilde{V}_1}{8} = \frac{\widetilde{V}_1}{-j4}$$
$$\frac{\widetilde{V}_2 - \widetilde{V}_1}{8} + \frac{\widetilde{V}_2 - \widetilde{V}_1}{j2} + \frac{\widetilde{V}_2 - 20e^{-j\frac{\pi}{3}}}{8} + \frac{\widetilde{V}_2(8 - j2)}{68} = 0$$

(d) Write the equations you derived in part (b) and (c) in a matrix form, i.e., $A \begin{bmatrix} \widetilde{V}_1 \\ \widetilde{V}_2 \end{bmatrix} = b$ Solve for A and b in numerical form.

Solution: From the above two equations, we have

$$A = \begin{bmatrix} \frac{1}{8} - j\frac{1}{4} & -\frac{1}{8} + j\frac{1}{2} \\ -\frac{1}{8} + j\frac{1}{2} & \frac{1}{4} + \frac{8}{68} - j\frac{1}{2} - j\frac{2}{68} \end{bmatrix} = \begin{bmatrix} 0.125 - j0.25 & -0.125 + j0.5 \\ -0.125 + j0.5 & 0.3676 + j0.5294 \end{bmatrix}$$
$$b = \begin{bmatrix} 0 \\ \frac{20e^{-j\frac{\pi}{3}}}{8} \end{bmatrix} = \begin{bmatrix} 0 \\ 1.25 - j2.165 \end{bmatrix}$$

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(e) Solve the systems of linear equations you derived in part (d) with any method you prefer, and then find $i_c(t)$.

Solution: 2 by 2 matrix is easy to be inverted, .i.e.,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3.1280 - j2.8782 & 1.5240 - j3.0381 \\ 1.5240 - j3.0381 & 1.1643 - j1.4291 \end{bmatrix}$$

With that we find

$$\begin{bmatrix} \widetilde{V}_1 \\ \widetilde{V}_2 \end{bmatrix} = A^{-1}b = \begin{bmatrix} -4.6726 - j7.0973 \\ -1.6388 - j4.3071 \end{bmatrix} = \begin{bmatrix} 8.4973e^{-j2.1530} \\ 4.6083e^{-j1.9344} \end{bmatrix}$$

Then

$$I_C = \frac{\widetilde{V}_1}{-j4} = \frac{j}{4}\widetilde{V}_1 = 2.1243e^{-j0.5822}$$

transform back to time domain we have

$$i_C(t) = 2.1243\cos(50t - 0.5822)$$

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